

“Urban highways quietly weaken social ties, new research shows”

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Highways are meant to connect cities, but new research shows they can also divide them socially. A study of 50 major U.S. cities finds that people living near highways are less likely to form social connections across them, especially in neighbourhoods shaped by historic segregation.

Urban highways are often celebrated as symbols of progress and economic growth. Yet a new study published in *Proceedings of the National Academy of Sciences (PNAS)* suggests that these large infrastructures come with a hidden social cost. Researchers from the IT University of Copenhagen and international collaborators show that highways act as powerful barriers to everyday social connections between people living nearby.

The team examined social connections in the 50 largest U.S. cities and compared them with the layout of urban highways. By looking at who is socially connected and where people live, they asked a simple question: are neighbours on opposite sides of a highway less likely to be connected than those on the same side?

The findings are strikingly consistent across all cities studied: people living on opposite sides of an urban highway are significantly less likely to be socially connected than those living on the same side.

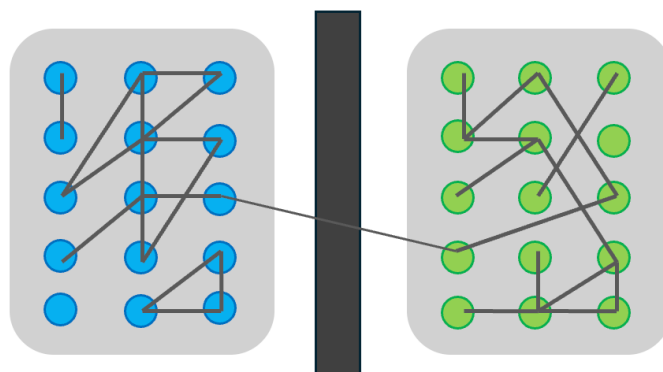


Figure 1: Simplified illustration showing how an urban highway can reduce social connections between nearby neighbourhoods.

“If you and I live on different sides of an urban highway, a barrier that we cannot cross on foot, it’s less likely that we will be friends than if we both live on the same side of it,” Vybornova explains. “This intuition has been there for many decades. The question was whether we could see it in the data and quantify it - and that’s what we managed to do”.

To measure this effect, the researchers developed a tool they call the *Barrier Score*. It shows how much highways reduce social connections between nearby residents. Across all 50 cities, highways were linked to fewer social ties, especially at short distances where people would normally meet on foot.

“What surprised us was not that we found an effect, but how consistent it was across all cities and distance ranges,” Vybornova notes.

The study also looks at the historical role of highways in the United States. Many urban highways were built in the mid-20th century in ways that divided or displaced Black neighbourhoods. The researchers find that these highways still act as strong social barriers today. “When you look at the history of highway construction in the U.S., it becomes clear that infrastructure was deliberately used to segregate communities,” Vybornova says.

Although the study uses data from online social networks, the findings point to real-world consequences. Everyday social ties, such as friendships and neighbourly relationships, are shaped by how easily people can reach one another. When infrastructure like highways makes these connections harder, their social impact can continue long after the roads are built.

The researchers also stress that the findings should not be read too simply. Highways can make long-distance travel easier, and the Barrier Score is not meant to decide on its own which roads should be changed or removed. “Urban planning is complex,” Vybornova says. “The Barrier Score can help inform decisions, but it’s only one piece of the puzzle.”

The findings suggest that roads do more than move traffic. They also shape how people meet, interact, and build communities. By making these effects measurable, the researchers hope to contribute to more socially aware urban planning.

About the researcher

Anastassia Vybornova is a Postdoctoral Researcher at the University of Copenhagen (KU) and a Visiting Researcher and External Lecturer at the IT University of Copenhagen (ITU). Her research focuses on urban inequality, social networks, spatial data science, and the social impacts of transportation infrastructure. She is a co-author of the study *Urban highways are barriers to social ties*, published in *Proceedings of the National Academy of Sciences* (2025).

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